NCBA Country Info Service — Kubernetes Troubleshooting Guide

Step 10 — Troubleshooting the application on Kubernetes

NCBA Integration Microservices

Kubernetes Troubleshooting Guide

This guide helps diagnose and resolve common runtime problems for the **Country Info microservice** running in the `ncba-countryinfo` namespace.

> All commands assume the namespace flag `-n ncba-countryinfo`. Set it once with `kubectl config set-context --current --namespace=ncba-countryinfo` to omit it.

---

0. First-line triage

Start every investigation with a broad snapshot, then drill in:

```
kubectl -n ncba-countryinfo get pods,svc,deploy,statefulset,hpa,ingress
kubectl -n ncba-countryinfo get events --sort-by=.lastTimestamp | tail -30
kubectl -n ncba-countryinfo describe pod <pod>          # 'Events:' section at the bottom
kubectl -n ncba-countryinfo logs <pod> --tail=200
kubectl -n ncba-countryinfo logs <pod> --previous       # logs from a crashed container
```

The application emits **structured JSON logs** (prod profile), so you can filter:

```
kubectl -n ncba-countryinfo logs deploy/countryinfo-app | grep '"level":"ERROR"'
```

---

1. Pod in `CrashLoopBackOff`

The container starts, exits, and Kubernetes keeps restarting it.

Diagnose

```
kubectl -n ncba-countryinfo describe pod <pod>          # look at Events + Last State (exit code)
kubectl -n ncba-countryinfo logs <pod> --previous       # the actual stack trace / startup error
```

Common causes & fixes

Symptom in logs	Cause	Fix
`Communications link failure` / `Unknown database`	DB not reachable at startup	Ensure MySQL pod is Ready; check `DB_HOST=mysql`, credentials in Secret
`Access denied for user`	Wrong DB credentials	Reconcile `secret.yaml` with what MySQL was initialised with
`OOMKilled` (in `describe`, Last State)	Heap exceeds memory limit	Raise memory limit, or lower `JAVA_OPTS` `MaxRAMPercentage`
`Port 8080 already in use`	Misconfiguration	Usually transient on restart; check no custom `SERVER_PORT` clash

> The liveness probe has a `startupProbe` guard (up to ~5 min) so a slow first boot is not mistaken for a crash. If a pod still crash-loops, it is a genuine startup failure — read `--previous` logs.

---

2. Pod stuck in `Pending`

The pod is scheduled but cannot start.

Diagnose

```
kubectl -n ncba-countryinfo describe pod <pod>          # reason is in Events
kubectl -n ncba-countryinfo get pvc                     # for the MySQL StatefulSet
kubectl get nodes -o wide
kubectl describe node <node> | grep -A5 Allocated
```

Common causes & fixes

Event message	Cause	Fix
`Insufficient cpu` / `Insufficient memory`	No node has the requested resources	Lower `resources.requests`, or add/scale nodes
`pod has unbound immediate PersistentVolumeClaims`	PVC cannot bind	Ensure a default `StorageClass` exists (`kubectl get storageclass`); on bare clusters install one (e.g. local-path)
`0/N nodes are available: node(s) had taint`	Taints/affinity	Add tolerations or deploy to a schedulable node

---

3. Service unreachable / no response

The app pods run, but requests to the Service time out or return nothing.

Diagnose

```
kubectl -n ncba-countryinfo get endpoints countryinfo-app    # MUST list pod IPs
kubectl -n ncba-countryinfo get pods --show-labels
kubectl -n ncba-countryinfo describe svc countryinfo-app
```

Checklist

- **Empty `ENDPOINTS`** → the Service selector (`app: countryinfo-app`) does not match pod labels, **or** no pod is *Ready* (only Ready pods are added to endpoints — check readiness probes).
- **Port mismatch** → Service `targetPort` must be `8080` (the container port); the Service `port` is `80`.
- **Test from inside the cluster** to isolate ingress vs. service issues:

  ```
  kubectl -n ncba-countryinfo run tmp --rm -it --image=busybox --restart=Never -- \
    wget -qO- http://countryinfo-app/actuator/health
  ```

---

4. Database connection refused

The app cannot reach MySQL.

Diagnose

```
kubectl -n ncba-countryinfo get pods -l app=mysql
kubectl -n ncba-countryinfo logs statefulset/mysql --tail=100
kubectl -n ncba-countryinfo get svc mysql                    # headless service must
        exist
kubectl -n ncba-countryinfo get configmap countryinfo-config -o
        jsonpath='{.data.DB_HOST}'; echo
kubectl -n ncba-countryinfo get secret countryinfo-secret -o
        jsonpath='{.data.DB_PASSWORD}' | base64 -d; echo
```

Checklist

- MySQL pod **Ready**? If `0/1`, read its logs — first boot initialises the DB and can take 30–60s; the app's startup probe tolerates this.
- `DB_HOST` in the ConfigMap must equal the MySQL Service name (`mysql`).
- Secret values (`DB_USERNAME`, `DB_PASSWORD`) must match the values MySQL was created with. If you changed them after first boot, the existing PVC still has the old credentials — recreate the DB (`delete pvc`) or fix the user.
- Connectivity test:

  ```
  kubectl -n ncba-countryinfo exec -it statefulset/mysql -- \
    mysql -u appuser -p"$DB_PASSWORD" -e "SELECT 1;"
  ```

---

5. High CPU / memory (and autoscaling)

Diagnose

```
kubectl -n ncba-countryinfo top pods                    # needs metrics-server
kubectl -n ncba-countryinfo get hpa countryinfo-app
kubectl -n ncba-countryinfo describe hpa countryinfo-app
```

Checklist

- **HPA shows `<unknown>/70%`** → metrics-server is not installed or not ready. Install it (`kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/...`) and confirm `kubectl top pods` works.
- **Pods pinned at the CPU limit (`500m`)** under load → the HPA should add replicas up to 10. If it is not scaling, check the HPA conditions in `describe hpa` (`AbleToScale`, `ScalingActive`).
- **Frequent OOM restarts** → raise the memory limit, or reduce JVM heap via `JAVA_OPTS` (`-XX:MaxRAMPercentage`).

- **External SOAP latency** inflating request threads → the SOAP client has 5s/10s timeouts + circuit breaker; inspect breaker state at `/actuator/health` and `resilience4j_*` metrics at `/actuator/prometheus`.

---

6. Rolling update stuck

A new revision will not finish rolling out.

Diagnose

```
kubectl -n ncba-countryinfo rollout status deployment/countryinfo-app
kubectl -n ncba-countryinfo get pods                    # new ReplicaSet pods not becoming
        Ready?
kubectl -n ncba-countryinfo describe pod <new-pod>      # readiness probe failing? image
        pull error?
```

Common causes & fixes

Cause	How to confirm	Fix
New image fails readiness	`describe pod` → readiness probe errors	Fix the image/config; rollout halts safely (`maxUnavailable: 0` keeps old pods serving)
`ImagePullBackOff`	`describe pod` Events	Wrong image name/tag or registry auth; fix `image:` / imagePullSecret
Bad config in new revision	new pods crash-loop	Roll back, then fix

Roll back to the last working revision:

```
kubectl -n ncba-countryinfo rollout undo deployment/countryinfo-app
kubectl -n ncba-countryinfo rollout history deployment/countryinfo-app
```

---

7. Ingress not routing

```
kubectl -n ncba-countryinfo describe ingress countryinfo-ingress
kubectl get pods -n ingress-nginx                       # controller installed & running?
```

- Confirm an ingress controller is installed and the `ingressClassName: nginx` matches it.

- Ensure the request `Host` header matches `countryinfo.local` (or your host), and that the host resolves to the ingress address (`/etc/hosts` or DNS).
- Test the Service directly (port-forward) to confirm the problem is the ingress layer, not the app.

---

8. Useful one-liners

```
# Tail only error logs across all app pods
kubectl -n ncba-countryinfo logs -l app=countryinfo-app --tail=-1 -f | grep ERROR

# Exec a shell into a running app pod
kubectl -n ncba-countryinfo exec -it deploy/countryinfo-app -- sh

# Watch pods during an incident
kubectl -n ncba-countryinfo get pods -w

# Recent events, newest last
kubectl -n ncba-countryinfo get events --sort-by=.lastTimestamp

# Restart the app cleanly (e.g. after a config change)
kubectl -n ncba-countryinfo rollout restart deployment/countryinfo-app
```

For deployment steps, see the companion **Kubernetes Deployment Guide**.